

Thermal performance of modified melamine foam/graphene/paraffin wax composite phase change materials for solar-thermal energy conversion and storage[☆]

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ABSTRACT

The development of phase change materials (PCMs) is hampered by issues like leakage, poor thermal conductivity, and poor light absorption. In this study, we innovatively combined modified melamine foam (MF) and graphene nanoparticle (GNP) to address these defects of PCMs in the solar-thermal energy system. The MF was modified to endow the solar-thermal energy conversion ability and superhydrophobic interface characteristics due to the polypyrrole (PPy) by the in-situ polymerization and the polymerized octadecylsiloxane (PODS) layer coating on the sponge skeleton. The composite PCMs consisting of the modified MF, paraffin wax (PW), and GNP were fabricated and exhibited good leakproof ability, high thermal conductivity, and excellent solar-thermal energy conversion capability. Results showed that due to the effects of the PPy and the PODS on the sponge skeleton structure, MF@PPy-PODS/PW composite PCMs showed a decrease in the mass loss ratio to less than 0.34% and an increase in the thermal conductivity of 0.33 W/m·K and the solar-thermal storage efficiency of 75.68%. Besides, adding GNP with higher concentrations impacted positively on the thermal conductivity and the solar-thermal energy conversion ability but negatively on the phase change enthalpies. MF@PPy-PODS/GNP3/PW composite PCMs increased the thermal conductivity to 0.59 W/m·K and the solar-thermal storage efficiency to 79.36% while decreasing the phase change enthalpy to 130.61 J/g. Furthermore, a thermoelectric conversion system driven by solar energy was developed to show the potential of composite PCMs for cleaner energy production. The open-circuit voltages of the MF@PPy-PODS/GNP3/PW composite PCMs achieved 0.78 and 0.91 V at 2 and 4 Sun, respectively. The open-circuit voltages remained for a period of time and slowly dropped without the simulated solar light irradiation. This study provided a potential strategy for the optimal performance of composite PCMs and their application in solar thermal systems.

1. Introduction

As the issue of reducing carbon emissions has become increasingly prominent, it is urgent to develop and use clean and sustainable renewable energy to replace fossil fuels in the future (Chu & Majumdar, 2012). As of now, solar energy is the most common and abundant source of renewable energy (Mathioulakis et al., 2017). Among the several forms to use solar energy, the solar-thermal energy conversion and storage is a typical renewable energy utilization technique, which has been employed in a wide range of applications in green buildings (Chaiyat, 2015), waste heat recovery system (Sharif et al., 2014), etc.

However, the instability of solar energy affects its application due to its low radiation energy density and intermittent nature (Zhang et al., 2016). Therefore, to obtain long-term solar energy utilization, designing a proper thermal energy storage system is essential for storing and supplying dispatchable thermal energy and improving the stability and conversion efficiency of the solar energy utilization system (Alva et al., 2017).

Latent heat thermal energy storage system depends on the melting and solidification process of phase change materials (PCMs) to store and release large thermal energy, allowing for the inter-regional and inter-temporal use of thermal energy (Kenisarin and Mahkamov, 2007). By

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combining the latent heat thermal energy storage system with the solar-thermal energy conversion technique, it is possible to effectively address the issue of solar instability (Alva et al., 2017). However, factors such as leakage, low thermal conductivity, and poor light absorption are fatal drawbacks of PCMs to hamper their application in the solar-thermal energy conversion and storage system (Tao et al., 2020). Therefore, the advancement of composite PCMs with good leakproof ability, high heat transfer performance, and efficient solar-thermal energy conversion and storage capability has received increasing attention.

To solve the leakage problem of liquid PCMs, the traditional method is to use physical encapsulation by loading PCMs into a container, while considering increasing the heat transfer surface-area-to-volume ratio, this method leads to a complex system design (Xu et al., 2022). Thanks to the rapid advancement of micro/nano-technology, more emphasis has been paid to encapsulating PCMs in micro/nano scale spaces to prepare stable-shaped PCMs (Cheng et al., 2021). The common encapsulation techniques mainly include two kinds: one is to form stable core-shell structures, such as micro/nano-encapsulation (Su et al., 2015), electrostatic spinning (Luo et al., 2013), etc.; the other one includes using the surface tension and capillary force of porous materials to physically attach PCMs, such as expanded graphite-based composite PCMs (Wang et al., 2020). The former encapsulation technology has a complicated preparation process and high cost and remains challenging in practical large-scale applications (Aftab et al., 2018). In contrast, the latter encapsulation technology has a simple preparation process and low cost and has broad application prospects in solving the leakage problem of liquid PCMs. However, the good compatibility of porous materials and PCMs should be guaranteed to ensure the leakproof capabilities of the encapsulation approach indicated above for physiorption of PCMs by porous materials (Mao et al., 2021). In particular, the wettability of the porous skeleton surface is the key to the penetration of PCMs into it. For example, the lipophilic reduced graphene oxide (rGO) is used to modify the skeleton of porous materials to reduce the leakage of the paraffin wax (PW) (Cheng et al., 2020). Besides, the rich interconnected structure of porous materials provides heat transfer channels in PCMs and accelerates the phase change process. This enhancement of composite PCMs helps to improve the heat transfer performance (Lin et al., 2020). Moreover, it is worth noting that the light absorption capacity of porous materials determines the solar-thermal energy conversion efficiency of the composite PCMs, due to the weak light absorption capacity of PCMs (Shao et al., 2021). Many researchers are devoted to improving the light absorption capacity of porous materials, and results are confirmed that coating various solar-thermal energy conversion enhancement fillers on the porous skeleton surface, including carbon nanotubes (Du et al., 2021), rGO (Yang et al., 2019), polypyrrole (PPy) (Zhou et al., 2013), etc., are beneficial to improve the solar-thermal energy conversion ability of PCMs. Therefore, excellent skeleton properties of porous materials facilitate meeting the requirements for good leakproof ability, high thermal conductivity, and high solar-thermal energy conversion efficiency of composite PCMs.

Aiming to remedy the lack of PCMs for solar-thermal storage utilization, adding nanoparticles is also an attractive strategy. It is demonstrated that the addition of nanoparticles to PCMs can increase the kinematic viscosity, thereby weakening the mobility of the liquid PCMs (Nikkhah et al., 2015), and improving the effective thermal conductivity of PCMs (Sajid and Ali, 2018). Furthermore, nanoparticles dispersed in PCMs exhibit significant light absorption enhancement performance (Yang et al., 2021), and thus PCMs achieve efficient solar-thermal energy conversion and storage capability, which is derived from the localized surface plasmon resonance (LSPR) effect of nanoparticles (Dong et al., 2022). Common nanoparticles with the LSPR effect include Au (Chen et al., 2017), Ag (Zhu et al., 2019), carbon nanotubes (Tang et al., 2014), BN (Yang et al., 2016), TiN (Ren and Yang, 2018), ZrC (Yang et al., 2019), etc.

Comprehensively, it is a promising choice to composite with porous

materials to overcome the drawbacks of PCMs. Besides, after the optimization by the nanoparticles, the leakproof ability, heat transfer performance, and solar-thermal energy conversion and storage capability of composite PCMs also achieve the desired improvement (Yang et al., 2021). However, previous work has focused more on the study of porous materials properties by the modification and optimization, and few studies improve the encapsulating and thermal properties of PCMs with porous materials and nanoparticles simultaneously for the application of the solar-thermal conversion and energy storage. Because the pore size of porous materials is in micro-nano scales, the addition of nanoparticles can supplement the thermal properties of PCMs in the micro-nano pores (Lin et al., 2020). Relying on the enhancement of porous materials and nanoparticles, combining these two strategies has the potential for further improving the solar-thermal energy conversion ability of composite PCMs.

To this end, this study innovatively combined modified porous materials and nanoparticles to address the defects of PCMs for solar-thermal energy conversion and storage. Firstly, melamine foam (MF) with the advantages of a three-dimensional continuous porous structure, high porosity, and good mechanical characteristics was selected as the porous material. The PPy by the in-situ polymerization and the polymerized octadecylsiloxane (PODS) layer coating on the sponge skeleton of MF were used to endow the MF with solar-thermal energy conversion capability and superhydrophobic interface characteristics (Zhou et al., 2013). Then, the PW with isothermal phase change property, high latent heat, good thermal stability, and low price was selected as PCMs (Cui et al., 2022). Besides, graphene nanoparticle (GNP) was selected to further improve the thermal performance of PCMs. Finally, composite PCMs consisting of the modified MF, PW, and GNP were prepared by the vacuum impregnation method (Milian and Ushak, 2019). The microstructures and thermophysical properties of the obtained composite PCMs were characterized. The addition of modified MF and GNP significantly improved the leakproof ability, thermal conductivity, and light absorption capability of composite PCMs. The solar-thermal storage efficiency of composite PCMs was investigated within a simulated light testing system. Based on the improved thermophysical properties and solar-thermal energy conversion and storage capability of composite PCMs, a thermoelectric conversion system with composite PCMs driven by solar energy was developed to achieve a solar to thermal to electrical energy utilization pathway for clean energy production. This study provided a novel strategy for improving the thermal performance of composite PCMs and their application in the solar-thermal conversion and storage system.

2. Experiment

2.1. Experimental materials

Commercial melamine foam (MF) was purchased from Beiyou New Material Co., Ltd. (Shanghai, China). Pyrrole (Py) monomer, polyethylene glycol 600 (PEG 600), Octadecyltrichlorosilane (OTS), and GNP were obtained from Aladdin Industrial Co., Ltd. (Shanghai, China). PW, anhydrous ethanol, ferric chloride hexahydrate ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$), and methylbenzene were obtained from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China).

2.2. Modified process of MF

The modified process of MF consists of two steps, as shown in Fig. 1. Firstly, MF modified by PPy was prepared based on the in-situ polymerization method (Sun et al., 2019). In detail, some cubic MF with the dimension of $20 \times 20 \times 20 \text{ mm}^3$ were soaked in the anhydrous ethanol and deionized water for a complete ultrasonically cleaning. After that, the cleaned MF was immersed into the based solution consisting of PEG 600, Py monomer, and deionized water, and then FeCl_3 aqueous solution was dropped slowly into the above mixture. The temperature for the

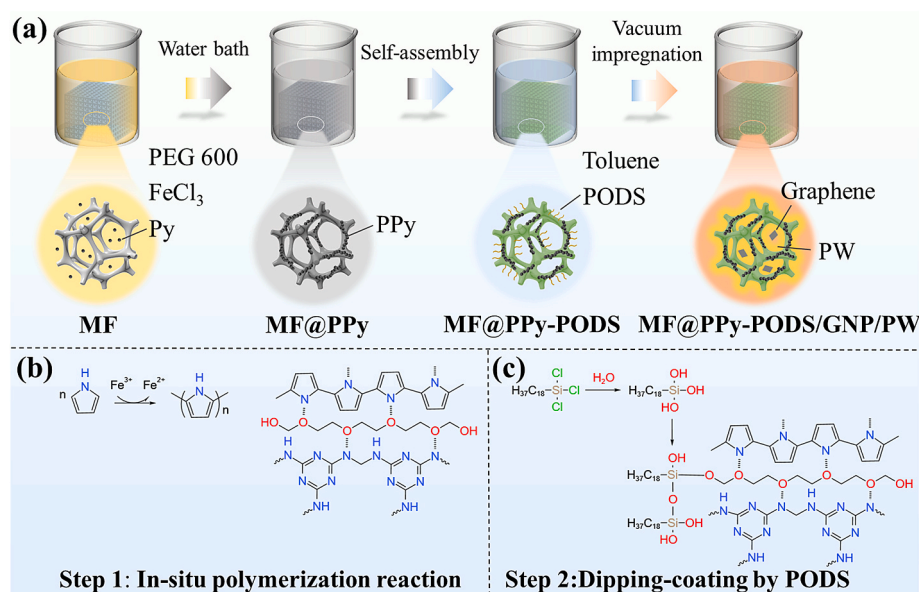


Fig. 1. (a) Schematic illustration for the preparation process of modified MF and composite PCMs. (b) In-situ polymerization reaction of PPy. (c) Dipping-coating by PODS.

in-situ polymerization reaction was set at 5 °C. To ensure sufficient Py monomers were provided during the reaction, the molar ratio of FeCl₃ aqueous solution and Py monomer was selected as 3:2, and the time of polymerization reaction was set as 5 h. In the chemical oxidation polymerization reaction, Fe³⁺ and PEG 600 were used as the oxidant and stabilizer to polymerize PPy from the Py monomer in an aqueous phase (Li, M. et al., 2014; Muller et al., 2013). As shown in Fig. 1(b), it was crucial in the polymerization reaction that the Py monomer was oxidized by Fe³⁺ to create a Py radical cation. As two Py radical cations dimerized, a bond that linked the two Py rings was formed by releasing H⁺. Afterward, the two Py rings were oxidized again and joined with Py radical cation intermediates to create tripolymers. Long-chain PPy molecules were created similarly by a continuous polymerization process (Li, X. et al., 2014). Subsequently, the obtained PPy was hydrogen-bonded to PEG, resulting in a PPy-PEG cluster with a complex cross-linked network structure caused by electrostatic interactions between the PPy and PEG molecules. Following that, plenty of cross-linking PPy-PEG clusters were firmly trapped on the sponge skeleton of MF through hydrogen bonding or electrostatic contact to change the MF from white to dark color. After the polymerization reaction, the samples were rinsed with anhydrous ethanol and deionized water to remove unreacted residues until the rinse solution turned colorless. Then the samples were dried in an oven at 100 °C again, and the prepared products were obtained, labeled as MF@PPy.

Secondly, the superhydrophobic modification of the MF@PPy was created by self-assembly of OTS to obtain the polymerized octadecylsiloxane (PODS) layer coating on the sponge skeleton (Zhou et al., 2013). 1 mL of OTS was dripped into 50 mL of toluene, and after magnetic stirring the mixture for 0.5 h, an OTS-toluene solution was obtained. MF@PPy was put into the above mixture solution and kept at room temperature for 1 h. Fig. 1(c) displayed the process of hydrophobic coating on the MF@PPy was formed in two steps, hydrolysis and condensation (Wang and Lieberman, 2003). To begin, the hydroxyl groups of the OTS were generated in situ by hydrolysis of the precursor. Then the generated hydroxyl groups proceeded to a reaction with the hydroxyl groups of the PEG and linked with the sponge skeleton of MF. The OTS hydrolyzed molecules responded synchronously by self-assembly cross-linking via Si–O–Si condensation. Consequently, the hydrophobic long-chain coating of PODS was immobilized on the sponge skeleton of MF. Following the modification, the samples were washed and dried again to get dark superhydrophobic products, termed

as MF@PPy-PODS.

2.3. Preparation process of composite PCMs

The preparation process of composite PCMs is shown in Fig. 1(a). In this study, PW was selected as the basic PCM. A certain amount of PW was put into a water bath at 80 °C and heated until fully melted. Different masses of GNP, as nanoparticle filler, were added to the molten PW for magnetic stirring and ultrasonic dispersion to prepare nanoparticle-phase change materials (nano-PCMs). Then the unmodified MF and modified MFs, including MF@PPy and MF@PPy-PODS, were dipped entirely into the above suspensions and impregnated for 1 h at 80 °C in a vacuum environment to remove air bubbles from the sponge skeleton. Once the vacuum impregnation process was complete, samples were taken out of the suspensions and cooled to room temperature. After that, samples were put into the oven to remove the residual PW from the surface of the composites, and the fabricated composite PCMs were obtained, termed as MF/PW, MF@PPy/PW, MF@PPy-PODS/PW, and MF@PPy-PODS/GNP/PW.

In this study, the components of the composite PCMs mainly include nano-PCMs and the sponge skeleton of MF, and the GNP concentrations are set as 0, 0.01, 0.05, and 0.1 wt%. According to the mass difference of samples before and after the vacuum impregnation process, mass ratios of nano-PCMs ($W_{\text{nano-PCMs}}$) and the sponge skeleton of MF (W_{MF}) in the composite PCMs are calculated and listed in Table 1.

2.4. Characterizations

The morphology and microstructure of the MF, MF@PPy, and MF@PPy-PODS, as well as composite PCMs, were described by scanning

Table 1
GNP concentrations, mass ratios of nano-PCMs and the sponge skeleton of MF in the composite PCMs.

Composite PCMs	C_{GNP} (wt%)	$W_{\text{nano-PCMs}}$ (wt%)	W_{MF} (wt%)
MF/PW	0	96.98	3.02
MF@PPy/PW	0	97.01	2.99
MF@PPy-PODS/PW	0	97.07	2.93
MF@PPy-PODS/GNP1/PW	0.01	97.16	2.84
MF@PPy-PODS/GNP2/PW	0.05	97.30	2.70
MF@PPy-PODS/GNP3/PW	0.1	97.51	2.49

electron microscope (SEM; Zeiss Supra 55, Germany, operating voltage: 15 kV). The chemical structures of PW, MF, MF@PPy, MF@PPy-PODS, and composite PCMs were characterized by Fourier transform infrared spectrometer (FTIR, Thermo iS20, USA, spectral range: 500–4000 cm^{-1}) and X-ray diffractometer (XRD, Bruker D8 Advance, Germany, scanning rate: 5°/min, operating voltage and current: 40 kV and 40 mA). The contact angles of MF, MF@PPy, and MF@PPy-PODS were measured by a contact angle instrument (DSA100S, Kruss, Germany).

A leakproof ability experiment was carried out on the encapsulation properties of PW and composite PCMs. The prepared samples were placed on filter paper and put at a constant temperature of 100 °C. The melting and leakage processes of PW and composite PCMs were observed simultaneously. The leakproof ability of composite PCMs was evaluated by comparing the wetted area of the filter paper and sample weight before and after heating, and the mass loss ratio (w) was calculated according to Eq. (1).

$$w = \frac{m_2 - m_1}{m_2} \times 100\% \quad (1)$$

where m_1 and m_2 were the mass of composite PCMs before and after the leakproof ability experiment.

The latent heat and phase change temperature of composite PCMs were analyzed using a differential scanning calorimeter (DSC, Discover DSC250, USA, temperature range: 20–80 °C, heating/cooling rate: 5 °C/min). And the thermal conductivity of composite PCMs was measured by a Hot Plate Thermal Constant Analyzer (TPS-2500S, Sweden). An infrared thermal camera (ThermoView TV40, Fluke, USA) was used to observe the temperature regulating capacity of composite PCMs. The samples were put on a thermostatic heating disk at 80 °C, and as the

heating period increased, the corresponding temperature distribution emerged. The light absorption characteristics of composite PCMs were characterized by an ultraviolet–visible–near infrared spectrophotometer (UV–Vis–NIR, Lambda950, PerkinElmer, USA).

A simulated light system was used to investigate the solar-thermal energy storage efficiency of composite PCMs. In this experiment, the simulant light source was provided by a xenon lamp (HSX-F300, NBeT, China). The irradiated intensity of the simulant light source reached 200 mW/cm^2 was assumed to 1 Sun. According to Eq. (2), the solar-thermal storage efficiency (η) was evaluated as follows:

$$\eta = \frac{m \cdot \Delta H}{S \cdot P \cdot t} \quad (2)$$

where m and S denote the mass and irradiated area of composite PCMs, ΔH indicates the melting enthalpy of composite PCMs, P represents the irradiated intensity of the simulant light source, and t is the irradiated time for the phase change process.

3. Results and discussion

3.1. Morphology and structure properties of modified MF and composite PCMs

Fig. 2(a–c) show the microstructures of MF, MF@PPy, and MF@PPy-PODS, respectively. As seen in Fig. 2(a), the rich three-dimensional interconnected pore structure of MF can be found with a pore diameter of 100 μm and a smooth surface (inset of Fig. 2(a)). A large number of three-dimensional connected porous skeletons in the MF provide highly ample space for filling the PCMs. Compared to the smooth surface

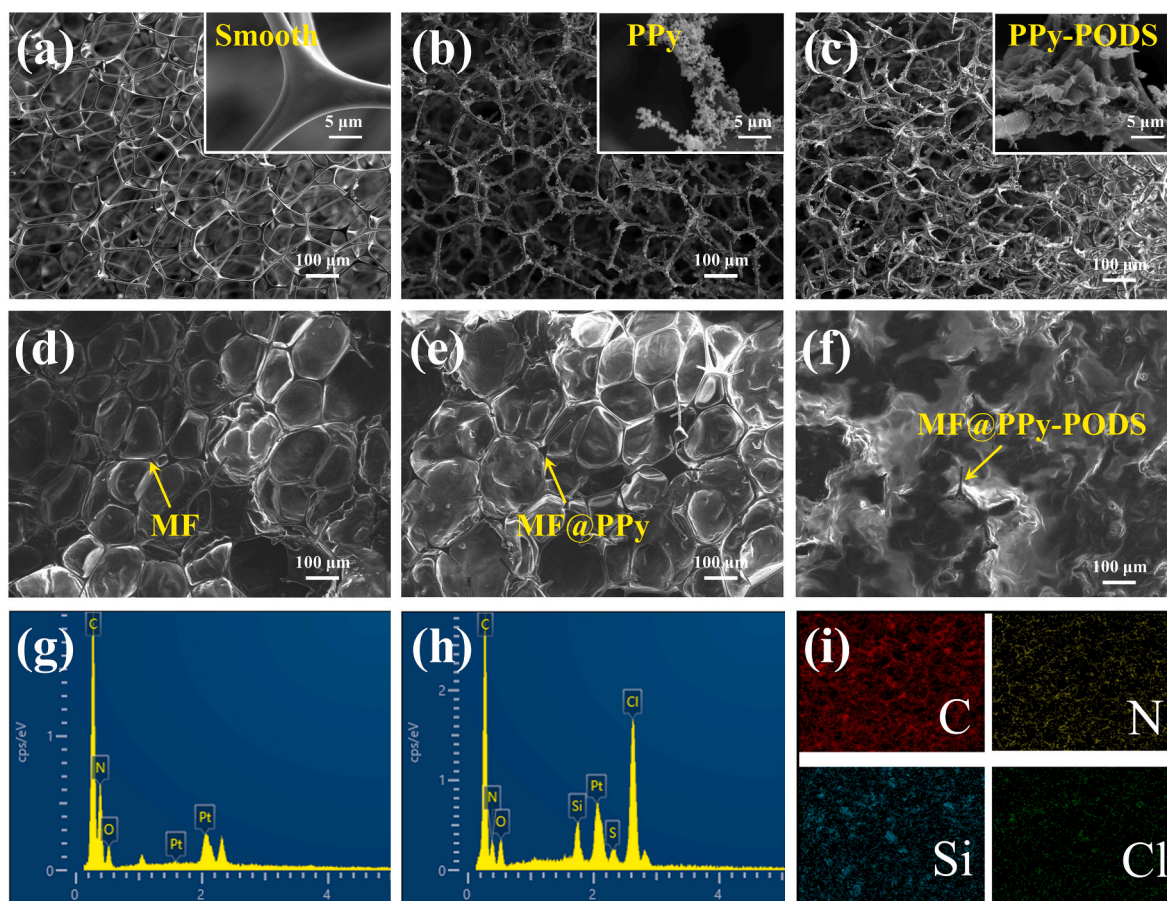


Fig. 2. SEM images of the (a) and (d) MF, (b) and (e) MF@PPy, (c) and (f) MF@PPy-PODS. EDS analysis of (g) MF and (h) MF@PPy-PODS. (i) C, N, Si, and Cl elemental mapping images of MF@PPy-PODS.

of the MF skeleton, Fig. 2(b) shows plenty of PPy nanoparticles loaded on the pore structure of MF after polymerization reaction, which results in a rough surface topographic feature of MF@PPy (inset of Fig. 2(b)). Furthermore, the surface profile of MF@PPy-PODS is illustrated in Fig. 2(c). After the self-assembly of OTS, the modified MF still maintains high absorbability features with the three-dimensional connected porous skeletons, which inherit from their parent MF. Notably, the PODS layer wraps on the skeleton structure of MF, which is the leading cause of the outstanding hydrophobicity of the sponge skeleton (inset of Fig. 2(c)).

Fig. 2(d–f) show the morphology properties of MF/PW, MF@PPy/PW, and MF@PPy-PODS/GNP/PW. After the vacuum treatment, the micropores of MF and MF@PPy are filled with the PW, while part of the sponge skeleton is still exposed outside the PW. But for MF@PPy-PODS, the sponge skeleton is entirely wrapped by the nano-PCMs, which shows a more satisfying impregnation performance. This is due to the sponge skeleton modified by PODS exhibiting better lipophilicity. The morphology of GNP is displayed in Fig. S1, and more details of the properties of GNP can be found in Supplementary Material.

Fig. 2(g) and (h) show the elemental composition of the MF and MF@PPy-PODS examined by EDS photoelectron spectroscopy. Compared to MF, MF@PPy-PODS appears with new peaks of Si and Cl, which can be explained as follows: firstly, the created PODS layer is coated on the sponge skeleton following the hydrolysis reaction of the chlorine group of the OTS to form hydroxyl groups (Kumar et al., 2015); secondly, after the in-situ polymerization reaction of PPy, a large amount of Cl^- from FeCl_2 remains in the solution. The residual Cl^- on the sponge skeleton may impact the peak of the element Cl (Yang et al., 2018). The peak of Pt is attributed to the spray pre-treatment of the sample before the analysis of the element. Moreover, the elemental mapping image in Fig. 2(i) proves that the elements C, N, Si, and Cl are homogeneously distributed throughout the image, indicating that the coating layer of PODS is uniformly covering the sponge skeleton of MF. It is evidenced that the MF@PPy-PODS is successfully prepared by the two modified steps based on the elemental analysis.

Fig. 3(a) and (b) show the FTIR measurement and XRD analysis to investigate the structures of MF, modified MF, PW, and composite PCMs. In the FTIR spectrum of MF, as displayed in Fig. 3(a), the absorption peaks at 3425 cm^{-1} , 1542 cm^{-1} , and 811 cm^{-1} are caused by the N–H and C=N stretching vibrations of MF, and the bending vibration of triazine ring, respectively (Wu et al., 2020). In the spectrum of MF@PPy, the characteristic peak at about 1465 cm^{-1} is attributed to the C–C stretching vibration of Py rings, indicating that the in-situ polymerization reaction of PPy does not act on the ring structure (Yang et al., 2018). The absorption peak at 1017 cm^{-1} corresponds to the C–H bond in the plane vibration of PPy. Comparing the spectra of MF and MF@PPy, it can be confirmed that PPy has been successfully prepared and attached to the sponge skeleton of MF. In the spectrum of MF@PPy-PODS, the new absorption peak at 1114 cm^{-1} is assigned to the Si–O–Si bond asymmetric stretching, suggesting the presence of PODS within the sponge skeleton (Ke et al., 2014). In the spectrum of PW, the absorption

peaks at 2916 cm^{-1} and 2848 cm^{-1} are caused by the $-\text{CH}_2$ asymmetrical stretching vibration of PW. The peaks at 1465 cm^{-1} , 1375 cm^{-1} , and 720 cm^{-1} are due to the $-\text{CH}_2$ and $-\text{CH}_3$ deformation vibration and the $-\text{CH}_2$ rocking vibration of PW, respectively (Wu et al., 2020). Compared to PW, the spectra of four composite PCMs are similar without new characteristic peaks, indicating that the immersion of PW and GNP into the sponge skeleton of MF and modified MF is a physical adsorption process.

To further characterize the structure properties, Fig. 3(b) shows the XRD patterns of PW, MF/PW, MF@PPy/PW, MF@PPy-PODS/PW, and MF@PPy-PODS/GNP3/PW composite PCMs. It can be observed that two diffraction peaks appear at 21.5° and 23.8° , which correspond to the (1 1 0) and (2 0 0) crystal planes of PW, respectively (Wu et al., 2020). Meanwhile, the composite PCMs show the diffraction peak positions consistent with PW, with no significant shift in peak position. The results still indicate that the crystal structure of PW and composite PCMs well intact without any chemical interaction during PW, GNP, and MF or modified MF. However, compared to PW, the peak values of the composite PCMs show a slight decrease due to the reduction of PW mass ratio in the composite PCMs caused by the addition of GNP and MF or modified MF (Shao et al., 2021).

3.2. Leakproof ability of composite PCMs

The remarkable elasticity and shape reversibility of the sponge skeleton can provide PCMs with a more flexible application. Benefiting from the three-dimensional cross-linked network structure of MF, the modified MF by the polymerization of PPy and coating of PODS still has excellent compression properties. As shown in Fig. 4(a–e), the MF@PPy-PODS is artificially compressed, bent, or twisted. Surprisingly, the MF@PPy-PODS can rapidly recover its original shape without significant structural fatigue, exhibiting excellent deformation resistance and good pressure reversibility during the deformation test experiment.

Additionally, the interface property of the sponge skeleton is related to the compatibility of PCMs impregnated into the MF. Fig. 4(f–h) show the contact angles of the MF, MF@PPy, and MF@PPy-PODS to characterize the wetting properties of their interfaces. The MF shows a contact angle of almost 0° in Fig. 4(f), and after the polymerization of PPy and coating of PODS, the contact angles of MF@PPy and MF@PPy-PODS are in the ranges of 57.8° – 59.6° (Fig. 4(g)) and 122.5° – 123.5° (Fig. 4(h)), respectively. To visualize the wettability difference of MF, MF@PPy, and MF@PPy-PODS, three samples mentioned above are put into the deionized water, and the result is recorded in Fig. 4(i). Samples of MF and MF@PPy are sunk in the deionized water, while the MF@PPy-PODS are floating on the deionized water. Furthermore, even if MF@PPy-PODS is pressed into the water by an external force, no water is impregnated into the sponge skeleton, and once the external force is removed, MF@PPy-PODS will float to the surface again. This demonstrates that the original MF and MF@PPy are hydrophilic, and MF@PPy-PODS is superhydrophobic under the effect of PODS wrapped on the

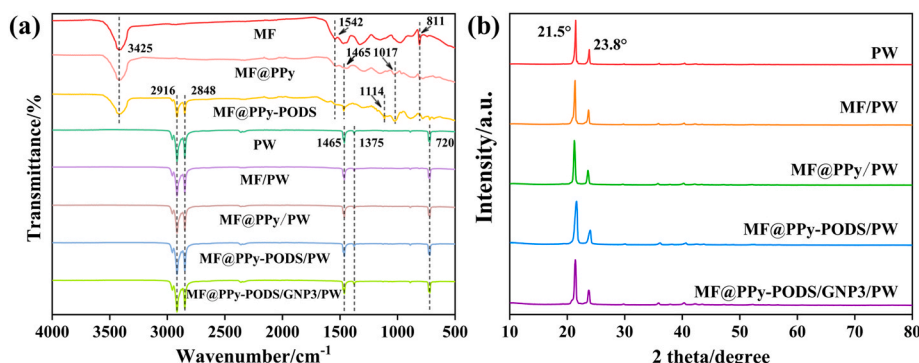


Fig. 3. (a) FT-IR spectra of MF, modified MF, PW, and composite PCMs. (b) XRD patterns of PW and composite PCMs.

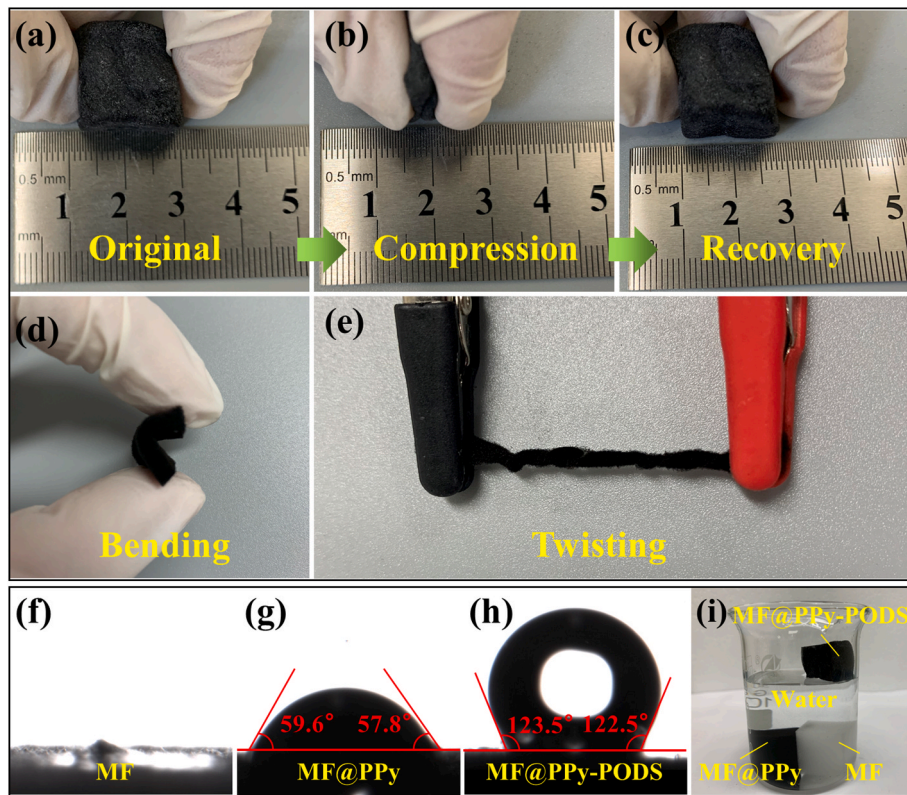


Fig. 4. Elasticity and shape reversibility test of MF@PPy-PODS, (a–c) compressing, (d) bending, and (e) twisting. Contact angles of (f) MF, (g) MF@PPy, and (h) MF@PPy-PODS. (i) Wettability test experiment.

sponge skeleton. According to the findings presented above, MF@PPy-PODS with excellent deformation resistance and superhydrophobic interface are prepared based on the MF as a template, endowing MF@PPy-PODS with unique functional capabilities.

The good compatibility of PCMs and the sponge skeleton of MF facilitates the absorption of PCMs and further prevents their leakage. The

leakproof ability of composite PCMs during the melting process is tested, as shown in Fig. 5. Fig. 5(a) exhibits photographs of leakage behavior composite PCMs at 100 °C for 0, 10, and 20 min, with the PW as the control group. The PW completely melted in 20 min and permeated into the filter paper due to the lack of leakage prevention strategy. For the MF/PW and MF@PPy/PW, the composite PCMs show a significant

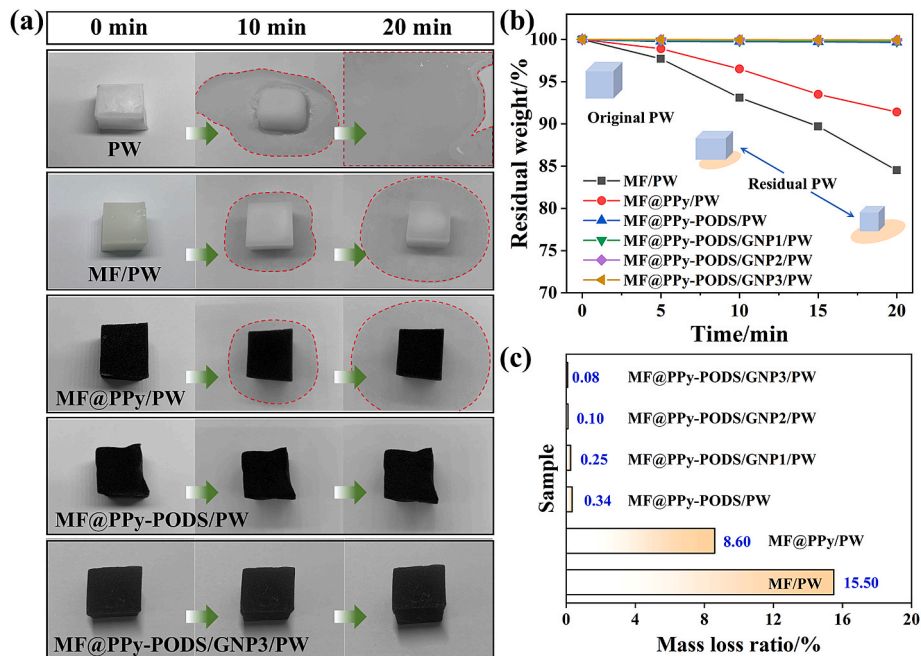


Fig. 5. (a) Leakage behavior for PW and composite PCMs at 100 °C for 0, 10, and 20 min. (b) Mass attenuation curves of composite PCMs at 100 °C. (c) Mass loss of composite PCMs at 100 °C for 20 min.

leakage in 20 min. However, the MF@PPy-PODS/PW and MF@PPy-PODS/GNP3/PW composite PCMs maintain stability without the significant leakage in 20 min. Calculated by Eq. (1), mass attenuation curves of composite PCMs at 100 °C within 20 min are presented in Fig. 5(b). The residual weight of PW in the MF/PW drops most rapidly with the heating time prolonging, followed by MF@PPy/PW. In contrast, the residual weight of PW in the MF@PPy-PODS/PW and MF@PPy-PODS/GNP3/PW has hardly changed. Besides, Fig. 5(c) shows the mass loss of the composite PCMs at 100 °C at the heating time of 20 min. The mass loss rates of PW in MF/PW, MF@PPy/PW, MF@PPy-PODS/PW and MF@PPy-PODS/GNP1, 2, 3/PW are 15.50%, 8.60%, 0.34%, 0.25%, 0.10%, and 0.08%, respectively. The explanation for this phenomenon is concluded to the following three reasons: the first is that the PPy attached to the MF reduces the interfacial energy between PW and the sponge skeleton and enhances the capillary effect of the sponge skeleton (Khosravi and Azizian, 2015); the second is that the PODS layer coating on the sponge skeleton is hydrophobic and facilitates improved compatibility between PW and modified MF, allowing PW to be more easily impregnated into the sponge skeleton (Zhang et al., 2017); and the third is that the addition of GNP weakens the fluidity of melted PW to a certain extent (Singh et al., 2020).

3.3. Thermophysical properties of composite PCMs

Phase change temperature and latent heat are among the most vital parameters for determining the thermal energy storage properties of PCMs, which can be analyzed by DSC measurement. Fig. 6 displays the phase change behaviors of the PW, MF/PW, MF@PPy/PW, MF@PPy-PODS/PW, and MF@PPy-PODS/GNP1, 2, 3/PW composite PCMs. As shown in Fig. 6(a) and (b), the upward and downward curves represent the endotherm and the exotherm effects, respectively. It can be noticed that the DSC curve of PW shows two phase change peaks, where the first peak represents the solid-solid phase change process with a temperature of approximately 33 °C; the second peak represents the solid-liquid phase change process with a temperature of approximately 53 °C. Furthermore, the DSC curves of composite PCMs are pretty similar to those of PW, especially phase change temperature. As illustrated in

Fig. 6(c), the addition of GNP and MF or modified MF shows a slight impact on the phase change temperature of composite PCMs, while it has a negative effect on the phase change enthalpies of composite PCMs. Taking the melting enthalpy as an example, compared with PW (152.47 J/g), the melting enthalpies of MF/PW (141.89 J/g), MF@PPy/PW (139.56 J/g), MF@PPy-PODS/PW (138.82 J/g) are relatively lower. It is apparent that the above three composite PCMs exhibit a favorable latent heat storage density due to the significant effect of PW. The reduction of melting enthalpy is attributed to the sponge skeleton occupying part of the space for PCMs. Besides, the melting enthalpies of MF@PPy-PODS/GNP1, 2, 3/PW composite PCMs (134.6, 132.47, and 130.61 J/g) further drop. It can be explained that the addition of GNP in the composite PCMs results in a decrease in the mass fraction of PW. Therefore, adding GNP with the appropriate mass fraction is essential to the residual latent heat of composite PCMs. Notably, similar phenomena of phase change temperatures and enthalpies are also found in the solidification process.

The cyclic thermal stability of composite PCMs guarantees their long-term service. Fig. 6(d) displays the DSC curves for MF@PPy-PODS/GNP3/PW before and after 100 heating/cooling cycles. It is found from the DSC curves that phase change temperatures and enthalpy of the MF@PPy-PODS/GNP3/PW remain almost unchanged after 100 cycles. The prepared composite PCMs show satisfied cyclic thermal stability, as evidenced by these reasonable and acceptable changes in temperature and latent heat in the long-term use process.

Thermal conductivity is an essential parameter in the thermal performance of composite PCMs. As shown in Fig. 7(a), the thermal conductivity of PW is undesirable at 0.28 W/m·K, which is comparable to Yang's study (Yang et al., 2019). However, as the PW is impregnated into the sponge skeleton of MF, the thermal conductivity value of MF/PW slightly decreases to 0.25 W/m·K. Two reasons can explain the decrease in thermal conductivity: the one is that the unmodified sponge skeleton of MF has a lower thermal conductivity, and the other is that the unmodified sponge skeleton of MF has the hydrophilic interface property with a more considerable interface thermal resistance. Conversely, Fig. 7(a) shows a rewarding improvement of the thermal conductivities of MF@PPy/PW and MF@PPy-PODS/PW to 0.31 and 0.33 W/m·K, which is attributed to the modification of PPy and PODS on

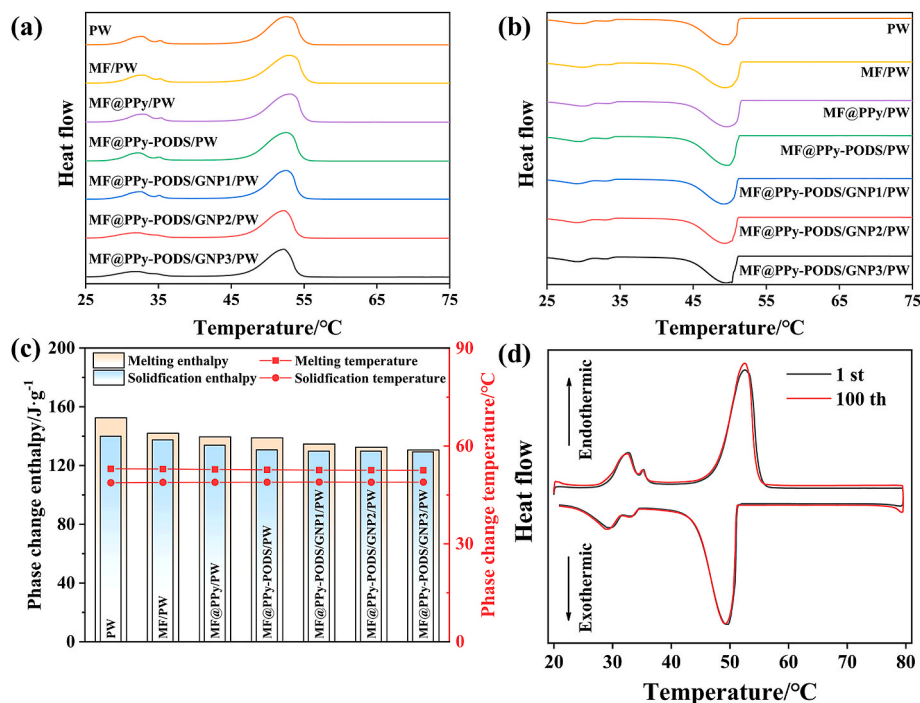


Fig. 6. Phase change behaviors of PW and composite PCMs. (a–b) DSC curves for melting and solidification processes. (c) Phase change temperature and enthalpy. (d) Cyclic thermal stability of MF@PPy-PODS/GNP3/PW before and after 100 heating/cooling cycles.

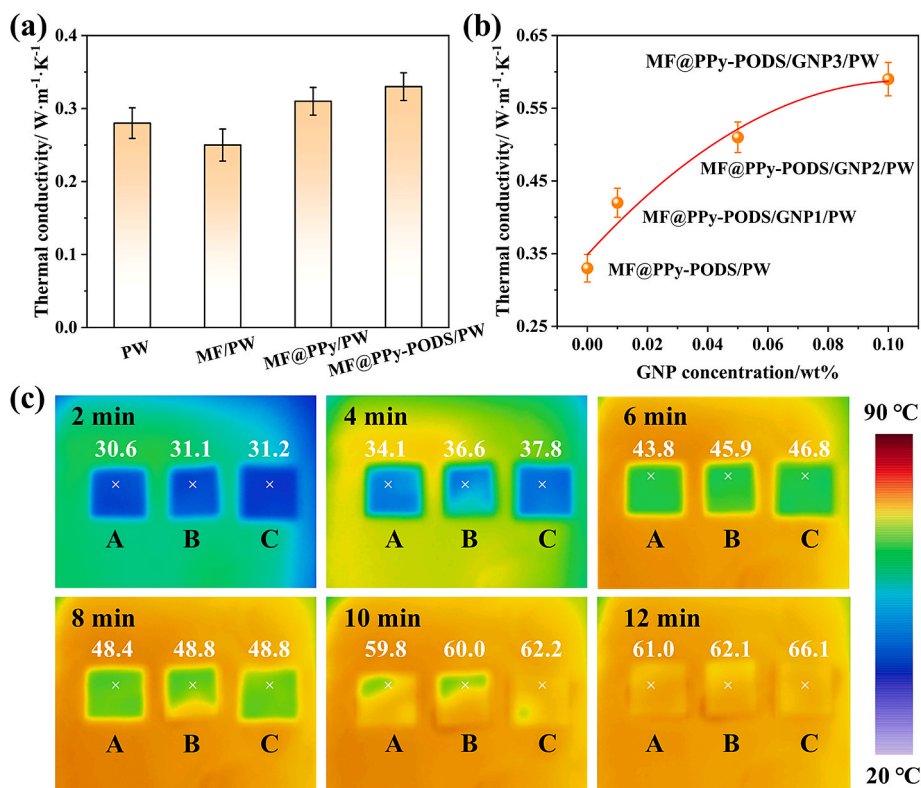


Fig. 7. (a) Thermal conductivity of PW, MF/PW, MF@PPy/PW, and MF@PPy-PODS/PW composite PCMs. (b) Thermal conductivity of the MF@PPy-PODS/GNP/PW composite PCMs with different GNP concentrations. (c) Infrared thermal images of the transient temperature distribution of MF/PW, MF@PPy-PODS/PW, and MF@PPy-PODS/GNP3/PW composite PCMs.

the sponge skeleton of MF. The thermal conductivity of the sponge skeleton is effectively improved by the polymerization of PPy. Furthermore, the interfacial lipophilic property of the sponge skeleton is enhanced by the coating of PODS, facilitating the reduction of the interfacial thermal resistance of the sponge skeleton. In addition, the thermal conductivities of MF@PPy-PODS/GNP/PW composite PCMs exhibit a significantly improving trend with the increasing GNP concentrations, as shown in Fig. 7(b). By comparison, the thermal conductivity of the MF@PPy-PODS/GNP3/PW composite PCMs rises to $0.59 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$, 110.71% and 78.79% higher than PW and MF@PPy-PODS/PW, respectively. That is, the thermal performance of the MF@PPy-PODS/GNP/PW composite PCMs is better than PW and MF@PPy-PODS/PW. As mentioned earlier, although the addition of GNP reduces the latent heat of composite PCMs, GNP can still be used as a candidate to improve the heat transfer of PCMs.

To further investigate the effect of latent heat and thermal conductivity on the temperature regulation capability of composite PCMs, Fig. 7(c) presents images of the transient temperature distribution during the melting process recorded by an infrared thermal camera. Samples A, B, and C with the same thickness of 5 mm represent MF/PW, MF@PPy, MF@PPy-PODS/PW, and MF@PPy-PODS/GNP3/PW, respectively. In the heating time of 2 min, the corresponding transient temperature of the samples is almost consistent, approaching 31°C . With the heating time prolonging, Sample C exhibits a slightly higher surface temperature than Samples A and B, and Sample C takes the lead in starting the solid-liquid phase change stage, followed by Samples B and A. In the heating time of 10 min, it is seen that the obvious color differences occur on the surfaces of three samples, suggesting that the PW in Sample C has melted completely, while a small amount of PW has not melted in Samples A and B. As the heating time increases to 12 min, three samples are in the liquid sensible thermal energy stage, and Sample C still shows a higher surface temperature than the other samples. The results confirm that MF@PPy-PODS/GNP3/PW exhibits the desirable heat transfer

performance as expected. Therefore, a reasonable combination of modified MF and GNP is an effective strategy to improve the thermal properties of composite PCMs.

3.4. Solar-thermal energy conversion and storage capability of composite PCMs

Solar-thermal energy conversion and storage is a meaningful way to utilize solar energy. Benefiting from the phase change process of PCMs to storing the thermal energy, the light absorption capacity of composite PCMs determines the solar-thermal storage efficiency. Fig. 8(a) shows the UV-Vis-NIR absorption spectra of PW and composite PCMs with a broadband absorption from 200 to 800 nm. The light absorption capacities of PW and PW/MF composite PCMs are relatively weaker. In contrast, after modifying the sponge skeleton of MF, the spectral absorption of MF@PPy/PW and MF@PPy-PODS/PW composite PCMs show a significant improvement. Besides, it is attractive that MF@PPy-PODS/GNP3/PW composite PCMs exhibit the strongest spectral absorption under the effect of GNP. This observation indicates that the modified MF is an excellent sponge skeleton for the solar-thermal energy conversion, imparting the composite PCMs with solar energy storage properties. Besides, GNP improves light absorption and solar-thermal energy conversion ability.

Fig. 8(b) shows the experimental device diagram, consisting of the simulant light source, thermal insulation material, and data acquisition unit. A thermal insulation layer covered the composite PCMs to reduce the heat loss. The thermocouple was put in the center of the composite PCMs to record temperature fluctuations. Fig. 8(c) displays the temperature changes of MF/PW, MF@PPy/PW, MF@PPy-PODS/PW, and MF@PPy-PODS/GNP3/PW composite PCMs during the solar-thermal energy conversion and storage processes. As the simulant light source turns on, the temperature of MF/PW rises slowly. Due to the poor light absorption ability of MF, the phase change process of MF/PW cannot be

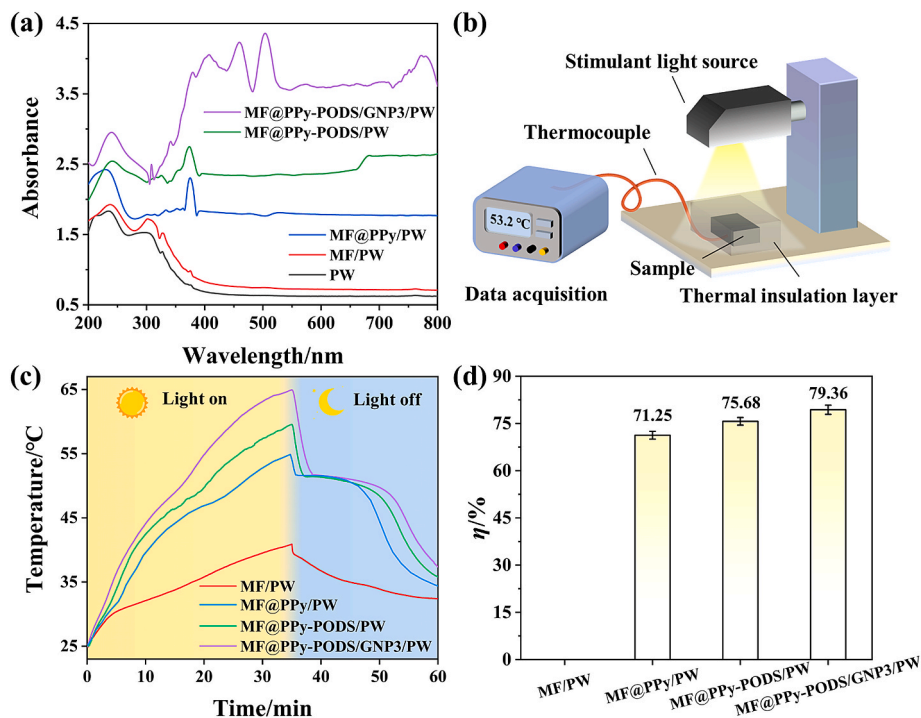


Fig. 8. (a) UV-Vis-NIR absorption spectra of PW and composite PCMs with a broadband absorption from 200 to 800 nm. (b) and (c) Experimental device diagram and the temperature curves for the solar-thermal energy conversion of composite PCMs. (d) Solar-thermal storage efficiency of composite PCMs.

completed under simulated solar light irradiation. Compared with the MF/PW, the temperatures of MF@PPy/PW, MF@PPy-PODS/PW, and MF@PPy-PODS/GNP3/PW composite PCMs rise rapidly, suggesting the good light absorption ability of these samples. When the temperatures reach the melting point of composite PCMs, slight temperature growth platforms between about 48 °C and 52 °C appear in these curves, indicating the melting processes of composite PCMs occur at this stage. When the melting processes are over, the temperatures rise rapidly again. It is noticed that among the above three composite PCMs with solar-thermal energy conversion ability, the melting time of MF@PPy-PODS/GNP3/PW is shorter than that of the other samples, and the temperature of MF@PPy-PODS/GNP3/PW during irradiation is higher than that of the other samples. This phenomenon is attributed to the high thermal conductivity and good solar-thermal energy conversion and storage ability of MF@PPy-PODS/GNP3/PW. As the stimulant light source turns off, the temperatures show a rapid decrease up to the solidification points of composite PCMs. The plateau stages occur around the solidification temperature, related to the solidification enthalpy release of composite PCMs. After that, the temperatures of the samples decrease again.

Based on the above Eq. (2), the solar-thermal storage efficiency of composite PCMs is illustrated in Fig. 8(d). As mentioned earlier, the MF/PW shows poor light absorption capacity without completing the melting process; hence, its solar-thermal storage efficiency is 0. A similar result is reported in the previous work (Cheng et al., 2020). However, the solar-thermal storage efficiencies of MF@PPy/PW and MF@PPy-PODS/PW composite PCMs show a remarkable increase, reaching 71.25% and 75.68%. Furthermore, this study indicates that MF@PPy-PODS/GNP3/PW composite PCMs exhibit the highest solar-thermal storage efficiency of 79.36% due to the solar-thermal conversion ability enhanced by GNP. This demonstrates that under the impact of the modified MF and GNP, composite PCMs show a satisfied solar-thermal conversion and storage capability.

3.5. Solar thermoelectric application

As shown in Fig. 9(a), the thermoelectric conversion system with composite PCMs driven by solar energy is proposed in this study, which consists of four parts: stimulant light source, MF/PW, and MF@PPy-PODS/GNP3/PW composite PCMs, thermoelectric device, and water cooling system. The thermoelectric device is used to generate electricity directly from a heat source (Borhani et al., 2021). To reduce the variation in power generation caused by temperature fluctuations on both sides of the thermoelectric device, the composite PCMs can be employed as the temperature stabilizer for the thermoelectric device to perform the solar-thermal conversion and storage process. The water cooling system maintains the cold side temperature of the thermoelectric device at 25 °C. Depending on this system, it is possible to convert solar energy into thermal energy and then into electric power.

As the lights turn on, the open-circuit voltage of the thermoelectric conversion system with MF@PPy-PODS/GNP3/PW reaches 0.78 V at 2 Sun, which is 1.59 times that of the MF/PW, as shown in Fig. 9(b). Due to the modified MF and GNP, MF@PPy-PODS/GNP3/PW has a better solar-thermal energy conversion and storage capacity than MF/PW. And the hot side of the thermoelectric device with the MF@PPy-PODS/GNP3/PW provides a higher temperature than the MF/PW. Furthermore, Fig. 9(c) displays that the open-circuit voltage of MF@PPy-PODS/GNP3/PW is improved remarkably at the light intensity of 4 Sun, reaching 0.91 V, which is 1.48 times that of the MF/PW. This phenomenon is attributed to the higher heat source temperature provided by the increased light intensity in the solar-driven thermoelectric conversion system. Interestingly, as the lights turn off, the voltages of two composite PCMs remain for a while and slowly drop. Indeed, the MF@PPy-PODS/GNP3/PW remains at a higher voltage than that of the MF/PW for the same time without the simulated solar light irradiation, indicating that the utilization of MF@PPy-PODS/GNP3/PW composite PCMs increases the output power and efficiency of the thermoelectric device and improves the sustainability of the system. This thermoelectric conversion system with composite PCMs driven by solar energy is distinguished by direct power production, high dependability, low

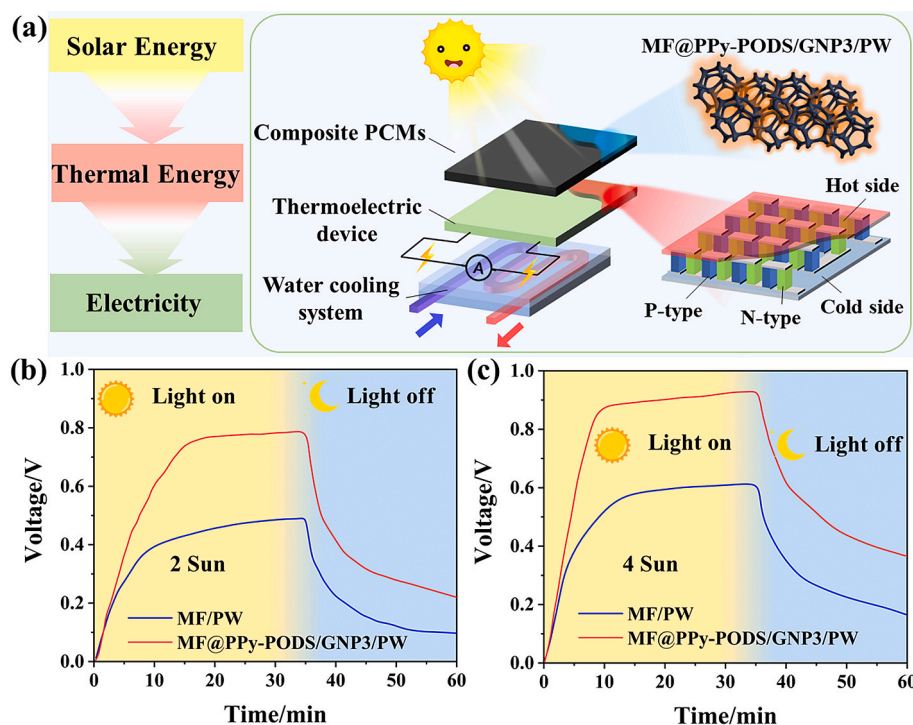


Fig. 9. (a) Experimental schematic diagram of the thermoelectric conversion system with composite PCMs driven by solar energy. (b) and (c) Open-circuit voltage of the thermoelectric conversion system with composite PCMs under the light intensity of 2 Sun and 4 Sun, respectively.

maintenance requirements, and environmental protection while also offering supportive evidence for the effective use of solar energy.

4. Conclusions

In this study, the superhydrophobic and flexible sponge skeleton of MF@PPy-PODS was facily devised and fabricated via the in-situ polymerization method of PPy and dipping coating of PODS. Then, the novel high-performance MF@PPy-PODS/GNP/PW composite PCMs consisting of the modified MF, PW, and GNP were successfully prepared by the vacuum impregnation method, integrating good leakproof ability, high thermal conductivity, and desired solar-thermal conversion efficiency. The major conclusions could be drawn as follows:

- (1) Due to the PPy and the PODS layer coating on the sponge skeleton, the modified MF ensured to provide the leakproof ability and improve the thermal conductivity and solar-thermal energy conversion efficiency of MF@PPy-PODS/PW composite PCMs. The mass loss rate of composite PCMs during the phase change process could be reduced to less than 0.34%. The thermal conductivity of MF@PPy-PODS/PW improved to 0.33 W/m·K. The solar-thermal storage efficiency of MF@PPy-PODS/PW composite PCMs reached 75.68%.
- (2) Increasing GNP concentrations exhibited an improving trend in the thermal conductivity and the solar-thermal energy conversion ability while showing a decreasing trend in the phase change enthalpies. Compared to PW, MF@PPy-PODS/GNP3/PW composite PCMs showed a 110.71% improvement in the thermal conductivity, increased to 0.59 W/m·K, and the highest solar-thermal storage efficiency, increased to 79.36%, while exhibited the lowest phase change enthalpy, decreased to 130.61 J/g. Therefore, adding GNP with an appropriate mass fraction was essential to the residual latent heat of composite PCMs.
- (3) As the temperature stabilizer in the thermoelectric conversion system driven by solar energy, the open-circuit voltage of the MF@PPy-PODS/GNP3/PW achieved 0.78 and 0.91 V at 2 Sun

and 4 Sun, which were 1.59 and 1.48 times that of the MF/PW. And the voltage of MF@PPy-PODS/GNP3/PW remained at a higher voltage than that of the MF/PW for the same time without the simulated solar light irradiation, owing to the effect of the modified sponge skeleton of MF and GNP on the output power and sustainability of the thermoelectric conversion system with composite PCMs driven by solar energy.

The prepared composite PCMs based on the modified MF and GNP showed good application prospects for solar-thermal energy storage. In addition, the thermoelectric conversion system with composite PCMs driven by solar energy had the advantages of direct power generation, high reliability, low maintenance requirements, and environmental protection. This study provided a potential strategy for developing and using clean energy.

CRediT authorship contribution statement

Wei Cui: Writing – original draft, Conceptualization, Methodology, Data curation. **Xiangxuan Li:** Investigation, Writing – review & editing. **Xinyi Li:** Methodology, Writing – review & editing. **Tianyu Si:** Writing – review & editing, Methodology. **Lin Lu:** Writing – review & editing. **Ting Ma:** Writing – review & editing. **Qiuwang Wang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2022.133031>.

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